This line, on extrapolation, cuts the 'X' axis at $\frac{-1}{K_m}$ and 'Y' axis at $\frac{1}{V_{max}}$.

The slope = $\frac{K_m}{V_{max}}$.

From these data both K_m and V_{max} can be determined (Figure 9.5)

For most of the enzymes, K_m lies between 10^{-1} and 10^{-7} M.

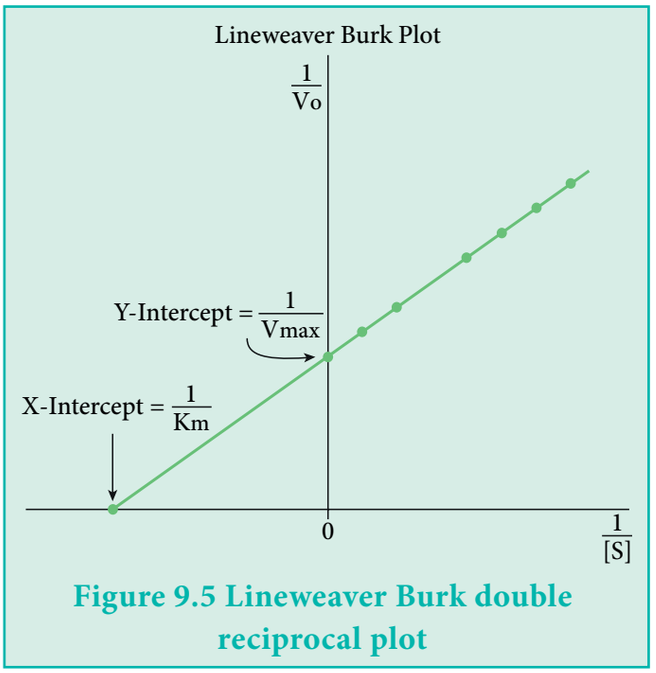

Figure 9.5 Lineweaver Burk double reciprocal plot

Significances of V_{max} :

1) V_{max} is the maximal velocity of the enzyme catalysed reaction.
2) This condition is attained when the catalytic sites on the enzyme are saturated with substrate. (i.e) when [S] >>> K_m.
3) $V_{max} = k_3$ [ET], where k_3 = rate constant for the dissociation of ES complex into product; [ET] = Total enzyme concentration.
4) k_3 = kcat = $\frac{V_{max}}{[ET]}$

The turnover number of an enzyme (kcat) is defined as the number of substrate molecules converted into product by an enzyme molecule in unit time when the enzyme is fully saturated with the substrate. The turnover numbers of most enzymes fall in the range of 1 to 10^4 sec^{-1}. The turnover number of different enzymes are given as follows :

S. NO.	ENZYME	TURNOVER NUMBER (SEC^{-1})
1	Carbonic anhydrase	6,00,000
2	Penicillinase	2,000
3	DNA polymerase I	15
4	Lysozyme	0.5

Haem is an iron-containing prosthetic group. It may function as an electron carrier and oxygen carrier in hemoglobin. It is also found in catalases and peroxidases, which catalyse the decomposition of hydrogen peroxide to water and oxygen.

XII U9 Enzyme Kinetics.indd 213 4/8/2020 3:27:13 PM